

$$T_{\text{model}}(t) = T_{\text{env}} + (T_0 - T_{\text{env}})e^{-kt},$$

$$r_i = T_{\text{obs},i} - T_{\text{model},i},$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |r_i|,$$

$$k_{\text{est}} = -\frac{1}{t} \ln \left(\frac{T(t) - T_{\text{env}}}{T_0 - T_{\text{env}}} \right) .$$